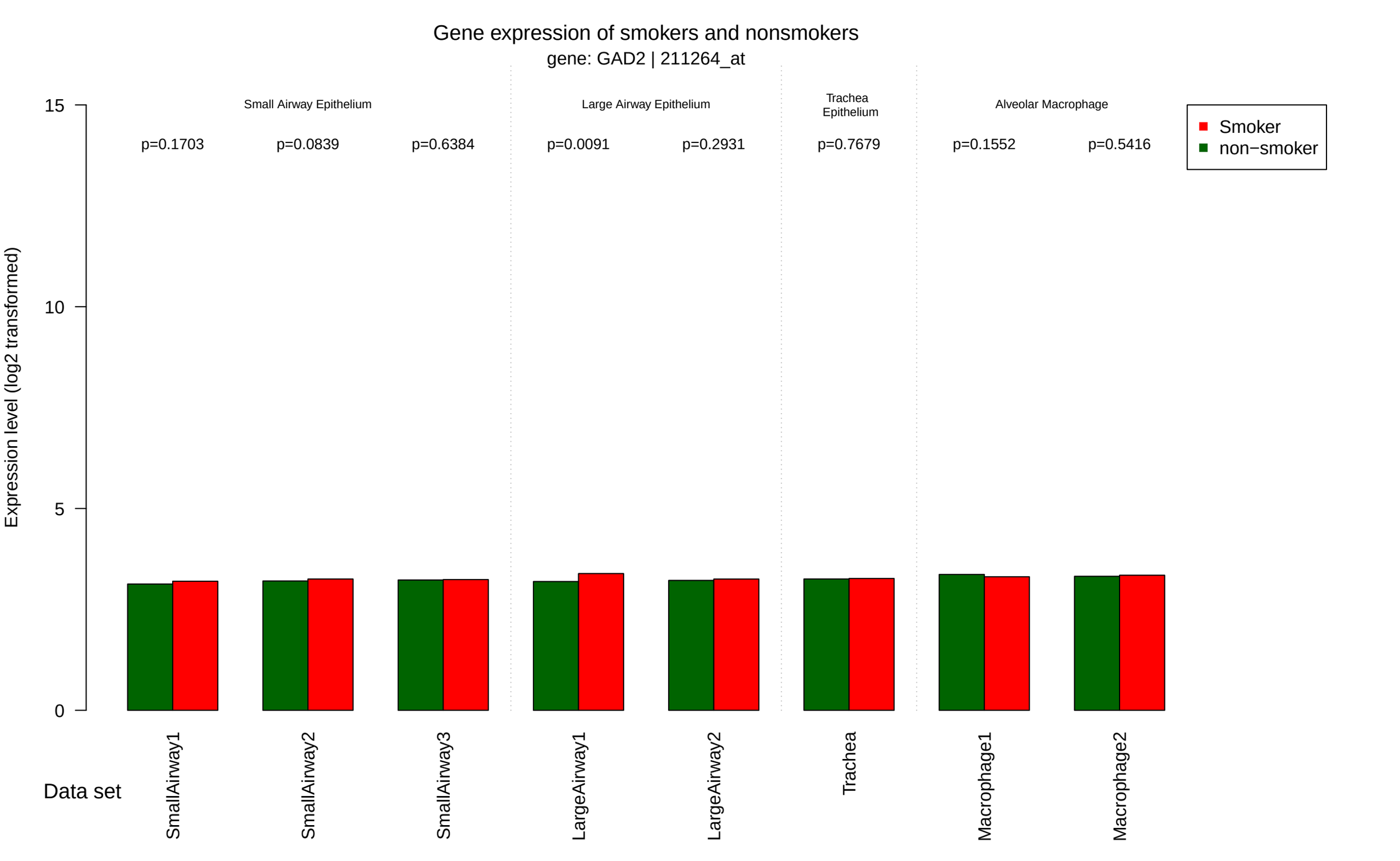
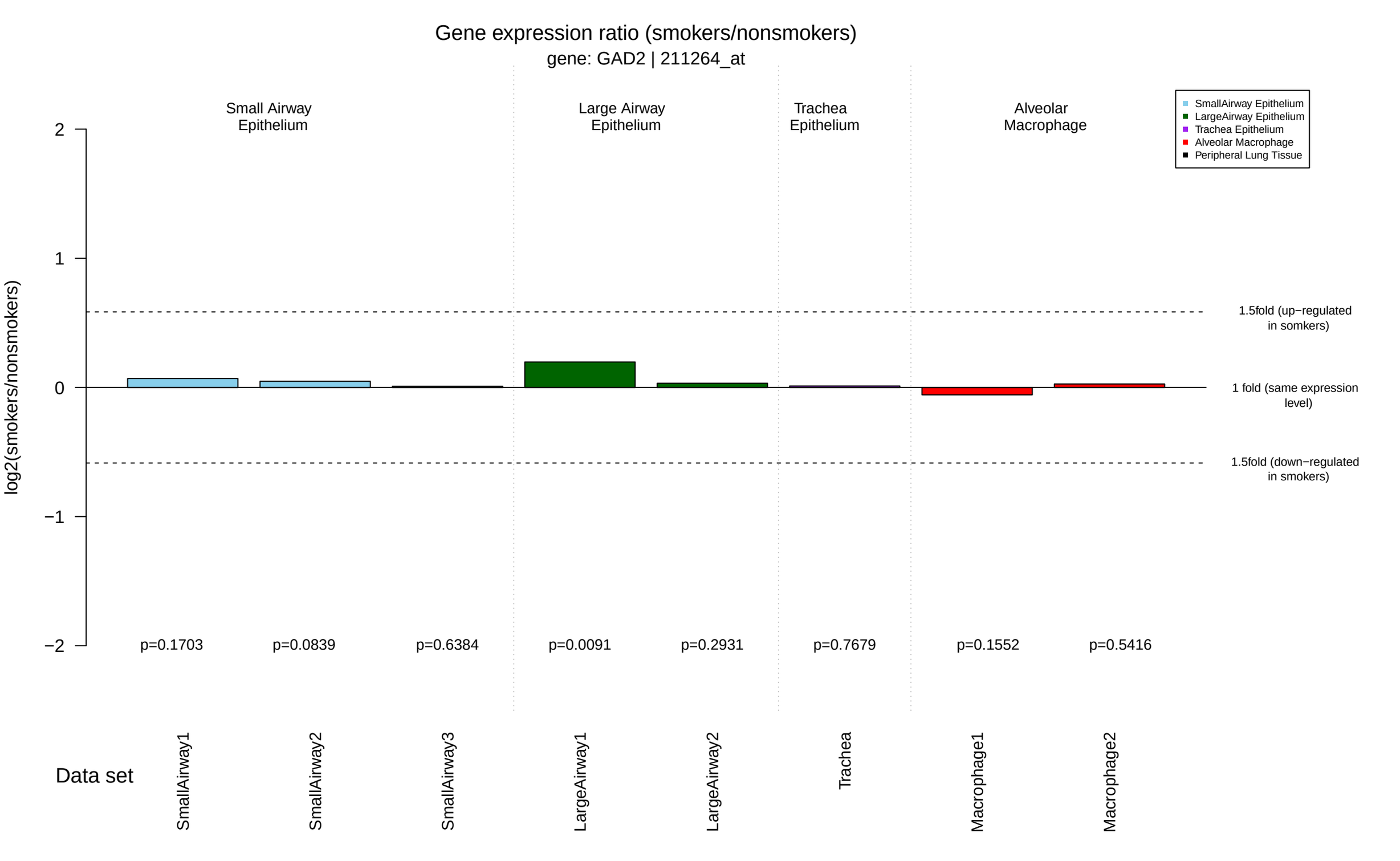
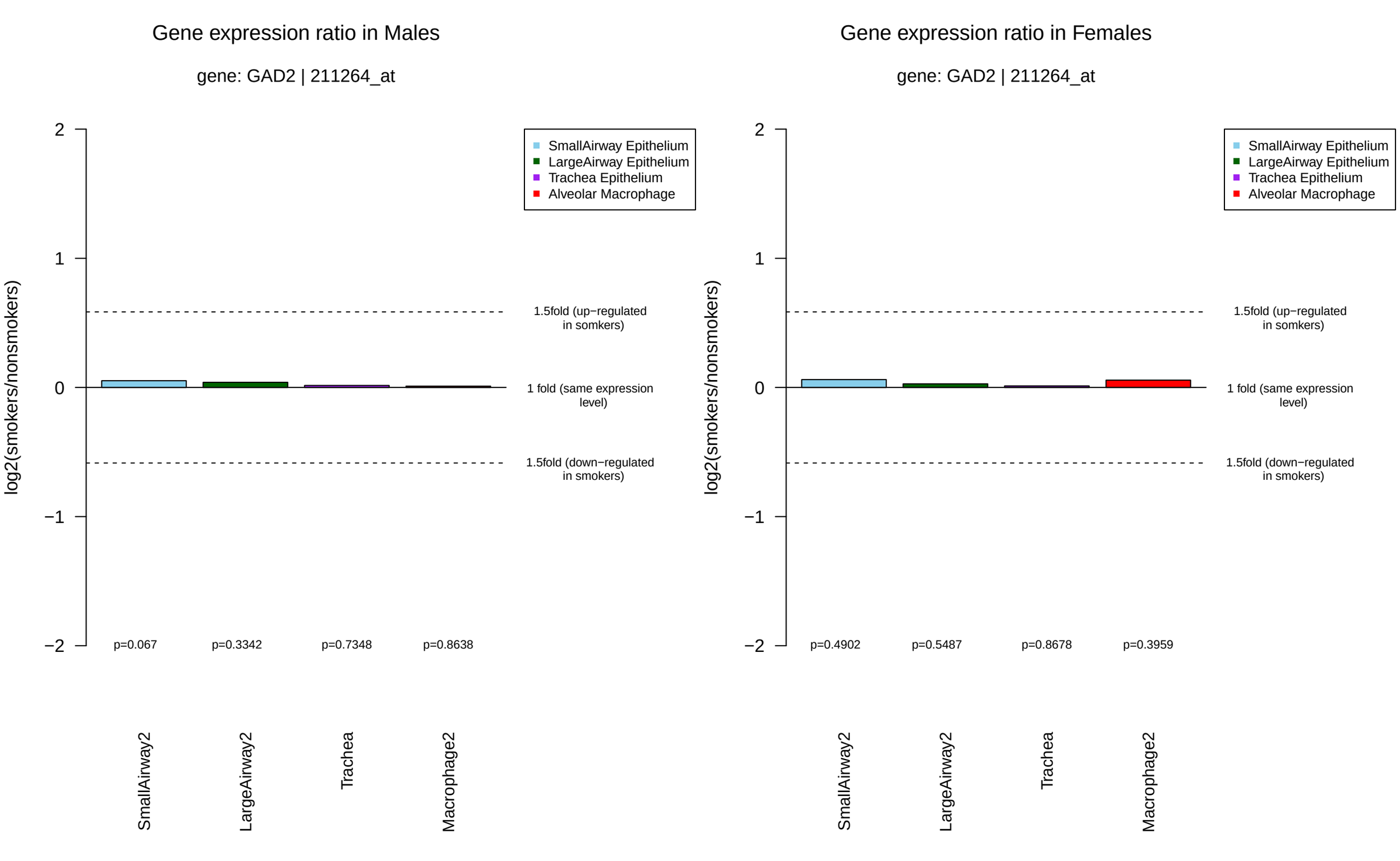
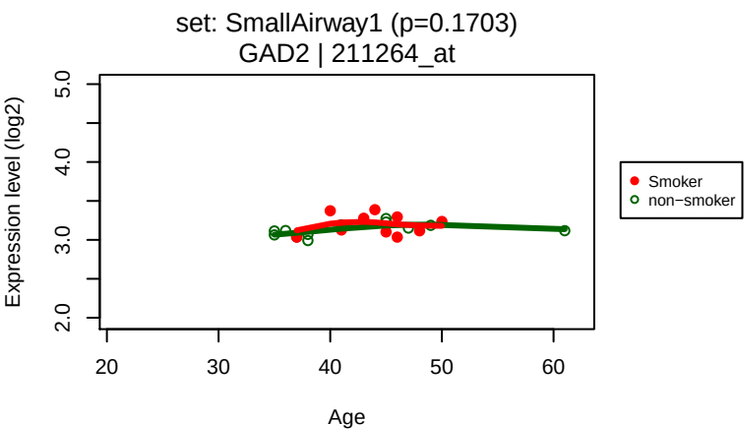
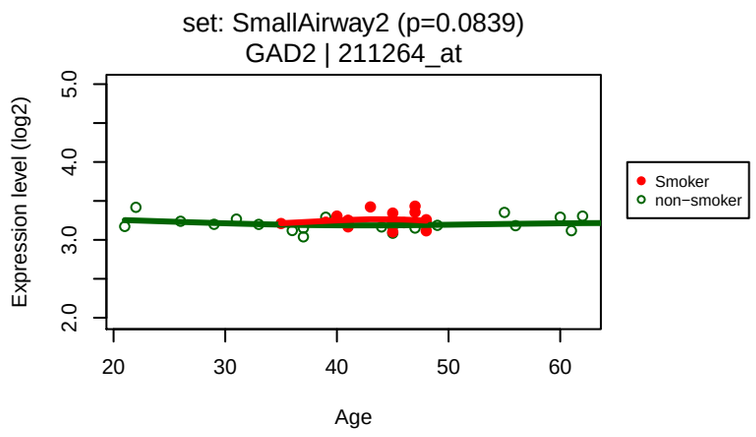
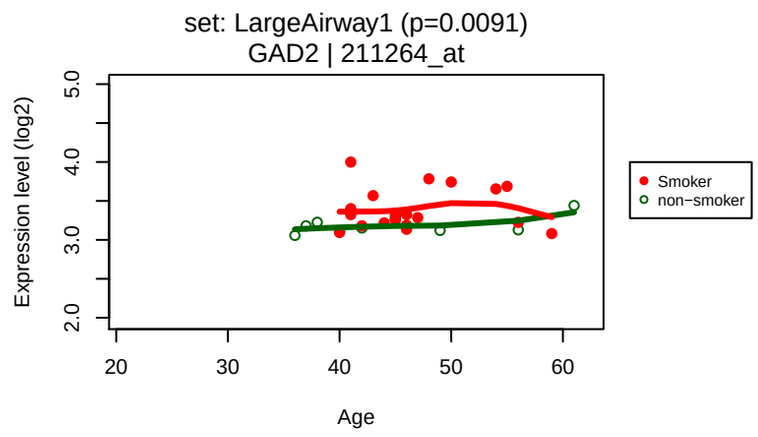

Gene expression of smokers and nonsmokers
gene: GAD2 | 211264_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1703
p=0.0839
p=0.6384
p=0.0091
p=0.2931
p=0.7679
p=0.1552
p=0.5416
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

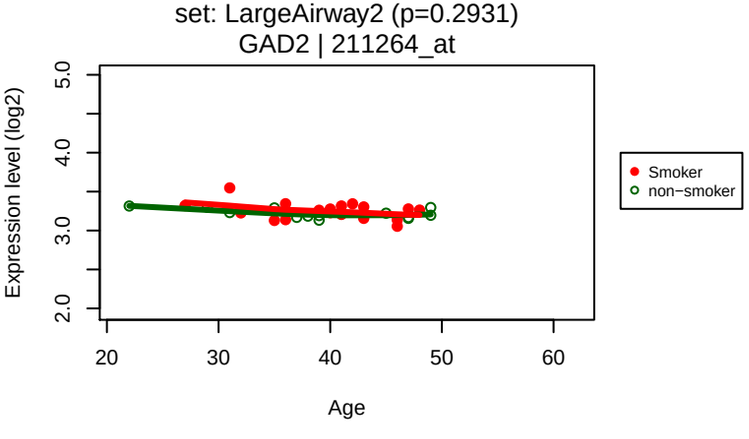

Gene expression ratio (smokers/nonsmokers)
gene: GAD2 | 211264_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1703
p=0.0839
p=0.6384
p=0.0091
p=0.2931
p=0.7679
p=0.1552
p=0.5416
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

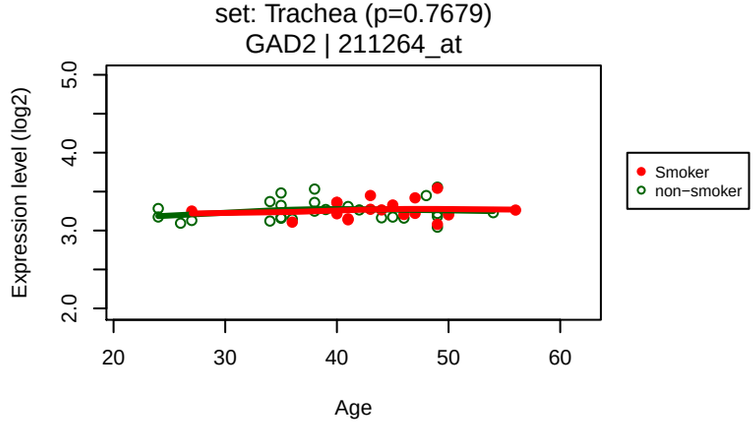

Gene expression ratio in Males
gene: GAD2 | 211264_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.067
p=0.3342
p=0.7348
p=0.8638
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GAD2 | 211264_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4902
p=0.5487
p=0.8678
p=0.3959
SmallAirway2
LargeAirway2
Trachea
Macrophage2

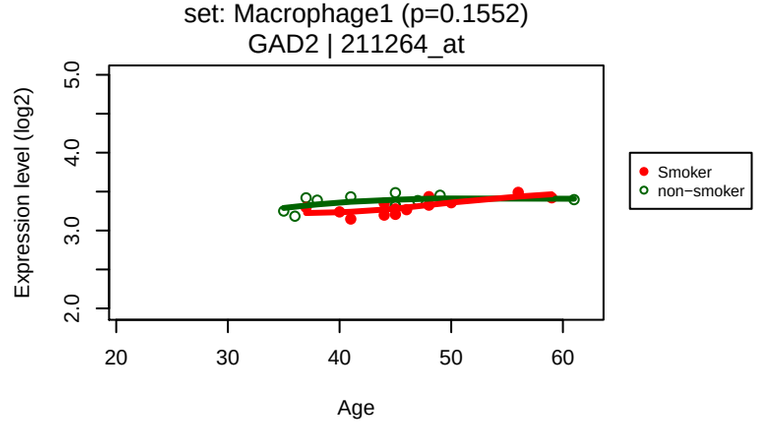

set: SmallAirway1 (p=0.1703)
GAD2 | 211264_at
Expression level (log2)
Age
Smoker
non−smoker

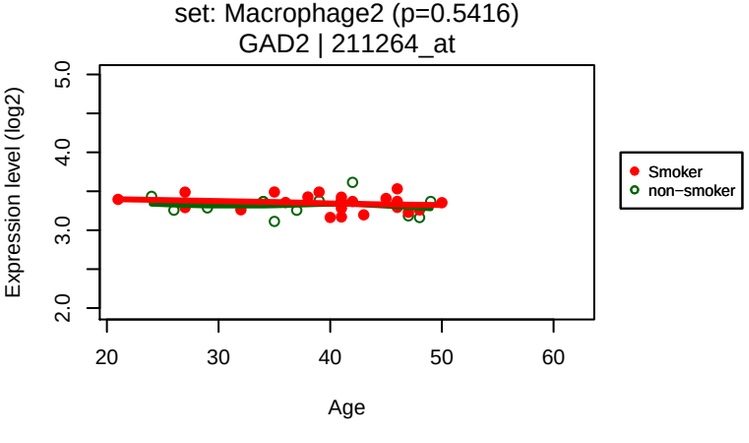

set: SmallAirway2 (p=0.0839)
GAD2 | 211264_at
Expression level (log2)
Age
Smoker
non−smoker

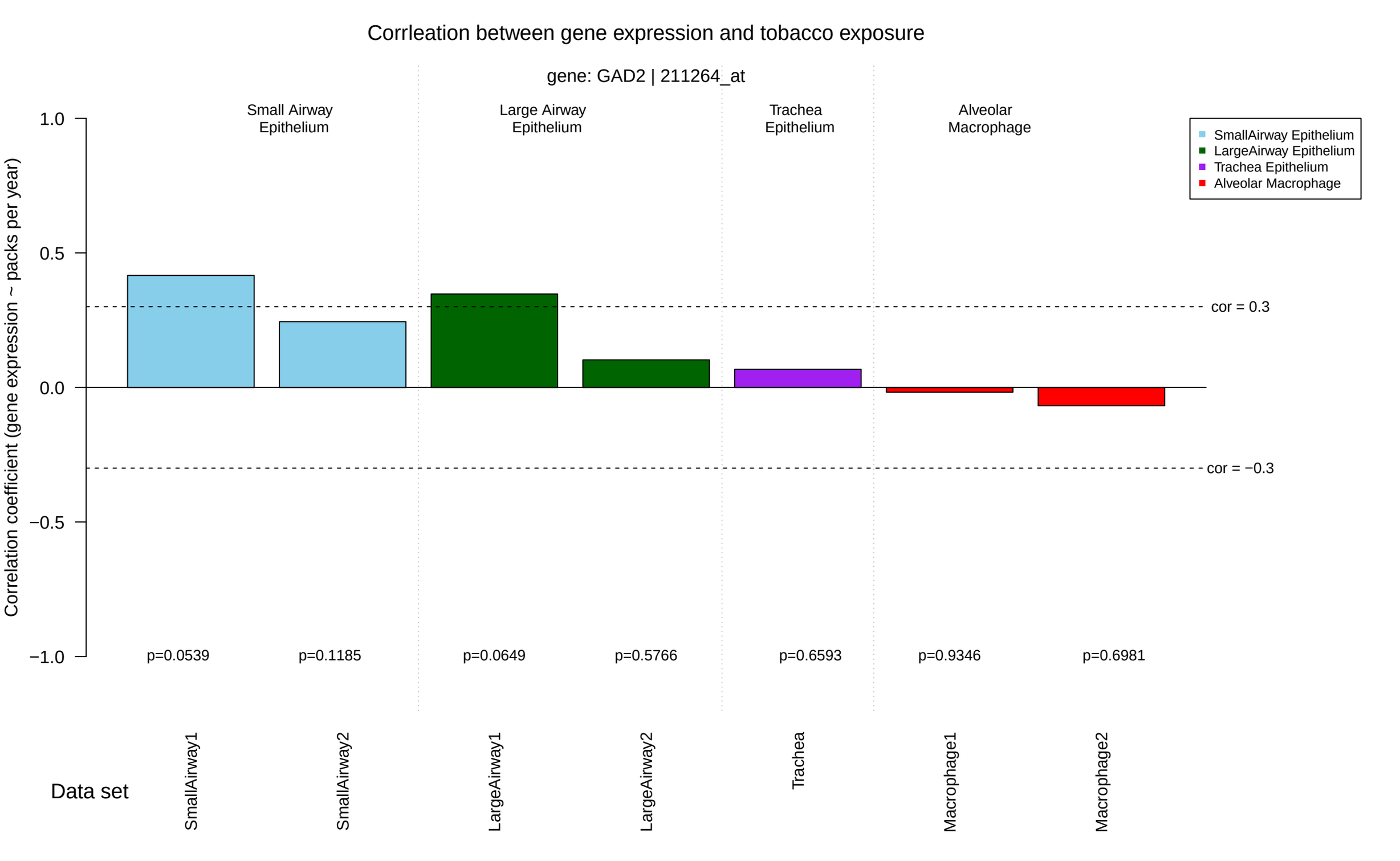

set: LargeAirway1 (p=0.0091)
GAD2 | 211264_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2931)
GAD2 | 211264_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7679)
GAD2 | 211264_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1552)
GAD2 | 211264_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5416)
GAD2 | 211264_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GAD2 | 211264_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0539
p=0.1185
p=0.0649
p=0.5766
p=0.6593
p=0.9346
p=0.6981
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2